

and variation across the sample of individuals. The wide bounds on the confidence intervals around the WTP measures indicate high variability across individuals in the sample, although the relatively small sample sizes are undoubtedly also a factor in the large confidence intervals. Nonetheless, these sources of randomness should be recognized when considering the difference between WTP and COI estimates.

Turning to the comparison of the two WTP point estimates, the confidence intervals overlap at even the 90% level of confidence, which would imply that the null hypothesis of equality between the two values cannot be rejected. However, this result should be confirmed with the complete combinatorial convolutions approach, as it has been shown to result in a lower likelihood of type II error than comparison of confidence intervals. This test results in a one-sided p -value of 0.31 and a two-sided p -value of 0.62. This confirms the comparison of confidence intervals, and we conclude that the null hypothesis of equality of the two WTP point estimates cannot be rejected at standard significance levels. It should be noted that the CVM WTP value is estimated as a discrete change and the DBM WTP value is estimated as a marginal change, which could create some disparity between the two.

VIII. CONCLUSIONS

There is considerable concern over the health effects individuals experience from exposure to the pollutants contained in wildfire smoke, and agencies such as the EPA often attempt to quantify the cost imposed on individuals as a result of this exposure. While it is recognized that estimates from a COI approach will often underestimate the economic value of reduced damages to human health, they will continue to be used if there are no other value estimates in the literature.

This study attempts to fill this gap by quantifying the individual value of a reduction in one wildfire smoke-induced symptom day by applying two nonmarket valuation approaches. Using data on the defensive actions individuals reported taking during California's Station Fire of 2009 along with their associated costs, the DBM application reveals

that individuals are willing to pay an average of \$86.87 for a reduction in one symptom day. The results of a CVM question reveal that individuals are willing to pay on average \$95.03 for a reduction in one symptom day. Comparing these values to a commonly monetized COI estimate reveals that for the case of wildfire smoke exposure, WTP values can be up to 30 times larger than a traditional COI estimate and up to five times larger than a more comprehensive COI estimate that incorporates the value of lost recreation. Analysis of confidence intervals reveals that while the mean CVM WTP value and marginal DBM WTP are statistically different from the COI estimates, they are not statistically different from one another. The complete combinatorial test further confirms this finding. This result is promising for future applications of both the CVM and DBM in the realm of valuation of health damages, as it provides a test of convergent validity between the two measures.

As this is the first study to estimate WTP values and a WTP:COI ratio for the specific case of reduced health damages from wildfire smoke exposure, these findings could be useful in a benefit transfer context. WTP values estimated in this study could be applied to future wildfires with similar characteristics to the Station Fire in terms of location, size, length of time until containment, effects on human health, and so on. However, care would have to be taken to ensure the accuracy of such a value transfer. The cost of human health damages from exposure to wildfire smoke is one component of the full cost of wildfire damages. Incorporating these monetized values into comprehensive cost estimates of a given wildfire can help inform objective analyses of the appropriate level of funding for fire prevention practices such as prescribed fire and forest thinning. In addition, if policy makers use COI estimates to quantify the cost of health damages from wildfire smoke, applying a conservative calibration factor of five found in this study may more accurately reflect the true value of these damages for the specific case of wildfire smoke exposure than applying ratios found in the literature specific to other pollutants.

Future studies should continue to compare WTP values to COI estimates for the specific case of wildfire smoke exposure to test whether the ratios estimated here are fairly consistent across wildfires. This could provide agencies with greater confirmation of the degree of understatement in benefit estimates typically associated with using a COI estimate. We also recommend that agencies incorporate less traditional costs associated with health symptoms such as the value of lost recreation when using a COI estimate, as this will bring the estimate closer to a more comprehensive welfare measure such as WTP. This may prove especially important for pollution exposure cases in which a large portion of individuals experience minor symptoms for which they do not seek medical attention. Future surveys could also question individuals about which specific recreation activities were missed as a result of symptoms in an effort to attach an accurate consumer surplus value to this loss. Further, collecting data on attitudes about the most important components of an individual's WTP for symptom day reduction could reveal whether defensive actions or disutility components of symptoms and lost leisure represent a significant economic cost to individuals.

Finally, two limitations to this study should be noted. First, we did not attempt to capture medical costs incurred by insurance providers. Second, while children are often heavily affected by wildfire smoke, it is difficult to apply revealed preference approaches such as the DBM to capture the economic value of reduced health damages imposed on children specifically. However, stated preference approaches could be used to question parents about their WTP to reduce the health effects experienced by children in the household specifically. This could be a valuable area for future research. This information will likely become increasingly important in areas such as California where large wildfires are moving closer to city centers and are no longer confined to rural areas.

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